

Comparison of K-Medoids and K-Means Algorithms in Segmenting Customers based on RFM Criteria

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Abstract. The company's approach to customers is important to maintain the company's profits. Understanding the differences of each customer is very important so that we can understand customer needs based on customer data. Customer Relationship Management (CRM) is considered as a solution to bridge the company and customers. Customer segmentation needs to be done to make it easier for companies to meet customer needs. Data mining and RFM modelling are used for customer segmentation in online retail companies using K-Means and K-Medoids methods. This research compares the performance of both algorithms using Davies Bouldin Index (DBI) and execution time. The results show K-Means is better in cluster validation and execution time. The average DBI value of K-Means is 0.2962 with an execution time of 0.0960 with k=3, while K-Medoids produces a DBI of 0.8942 and an execution time of 2.4295 with k=5. K-Means RFM customer tiers 1-3: Potential Customer/Golden Customer, Lost Customer/ Dormant Customer, and Superstar/Core Customer, 1-5: Champion and Lost. K-Medoids RFM 1-5: Lost, Loyal Customer, Champion, At Risk, and Hibernating, 1-3: Lost Customer/Dormant Customer, Potential Customer/Golden Customer, Superstar/Core Customer, Potential Customer/Golden Customer, and At Risk Customers/Occasional customer.

1 Introduction

In the digital era, the online retail business industry is experiencing rapid growth [1]. However, online retail companies are faced with great challenges in understanding and analyzing their customer behavior. This is due to the continuous increase in data volume and the complexity of diverse purchasing patterns [1]. To compete effectively and achieve success amidst fierce competition, a deep understanding of customer characteristics and preferences is crucial [2].

Recognizing the importance of the right approach to customers, companies need to focus on maintaining their profits [3]. Therefore, it is necessary to deeply understand the differences between each customer and their needs based on existing data [4]. One of the marketing methods used to manage all customers is Customer Relationship Management (CRM) [5], [6]. CRM helps companies collect and record all contacts between customers and companies [5], [6].

In an effort to understand customer needs based on customer data [3], companies segment customers using the unique characteristics of each customer [4]. This customer segmentation process is supported by data mining and RFM models [7].

The RFM model identifies customer groups with special treatment [1]. The RFM model is one of the most common segmentation methods that can identify the value of customers in the company with 3 variables: recency, frequency and monetary [8]. While data mining is a process in which valuable information is extracted from data sets through statistical analysis, data collection, and pattern analysis [9].

Data mining and clustering approaches enable the identification of customer segments based on complex RFM characteristics and patterns that may be difficult to see with manual processing alone [10]. Clustering is a data mining method where the way it works is by grouping data that has similar characteristics between one and the other data collected [9]. In this clustering method there are several algorithms that can be used such as: K-Means, K-Medoids, Fuzzy C-Means, DBSCAN, and so on.

K-Means and K-Medoids are commonly used in customer segmentation analysis as they can be used to group customers based on their purchasing patterns [11]. K-Means is a cluster analysis to divide objects into groups based on the nearest mean [12]. Whereas K-Medoids is a method that is an extension of K-Means, where K-Medoids is specifically designed to overcome the weakness of K-Means which is vulnerable to the influence of outlier data [1][13].

A commonly used performance evaluation is the Davies-Bouldin Index (DBI) [12]. In addition, execution time is used as an indicator to evaluate the efficiency and speed of an algorithm [14]. Lower execution time measurements indicate the algorithm's ability to process data quickly and responsively, which has implications for saving computational resources and enabling real-time data processing [14], [15].

Previous research shows K-Means has the best validity rate [16]. However, another study found K-Medoids to be better in some cases [1][2].

This research compares K-Means and K-Medoids in customer segmentation using RFM model with DBI and execution time as evaluation metrics. In addition, this study will also compare the interpretation of cluster results from both algorithms. Customer segmentation results will be evaluated based on the similarity of characteristics and different needs between customer groups generated by K-Means and K-Medoids.

In the segmentation of this study, a score range from 1 to 3 as well as 1 to 5 for each RFM variable will be used. The selection of these ranges is based on the consideration that a score range of 1 to 3 will help describe a more limited customer type, while a score range of 1 to 5 will provide more depth in the analysis, enabling deeper identification in customer behavior [17], [18]. The selection of this score range is expected to provide deep and thorough insights into the characteristics of different customers, thereby supporting smarter business decision-making.

2 Materials and Methods

2.1 Materials

The data used in this study comes from the Kaggle online repository. The data link used is <https://www.kaggle.com/code/hellbuoy/online-retail-k-means-hierarchical-Clustering/>. The dataset contains online transaction data from online retail stores and will be the data set used in this research. Then, the data is analyzed and clustered using two clustering methods, namely K-Means and K-Medoids.

2.2 Method

The research followed a six-stage workflow as shown in Fig. 1. The first stage is data pre-processing, where data exploration analysis, data transformation to obtain recency, frequency, and monetary variables, and data normalization are performed. After that, the data clustering stage is carried out using the K-Means and K-Medoids methods, followed by the evaluation of cluster validity with respect to execution time and the Davies-Bouldin Index method. After the cluster evaluation stage is completed, a score is assigned to each variable. The final stage involves analysis of the scoring results, customer segmentation, and interpretation of the results obtained.

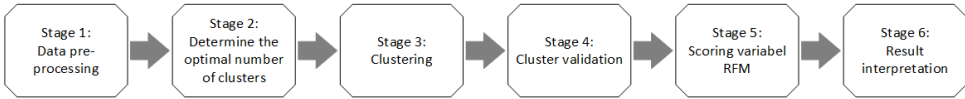


Fig. 1. Six stages of research workflow

2.2.1 The Optimal Number of Clusters

Before clustering, the optimal number of clusters is found using the elbow method. The elbow method selects the optimal number of clusters by considering the Sum Square Error (SSE) value in the cluster [19]. In the elbow method, SSE is calculated for each value of k (number of clusters) and the results are displayed in graphical form. The optimal number of clusters is determined by the point where an elbow is formed on the graph, indicating a low SSE value [20]. The SSE value is the sum of the average squared Euclidean distances between each point in the cluster and its centroid [3]. The formula for SSE is as follows [21]:

$$SSE = \sum_{k=0}^k \sum_{Xi \in S_k} ||Xi - Ck||^2 \quad (1)$$

Where k = number of clusters, xi = The i -th data, and Ck = cluster center point.

2.2.2 Clustering Method

2.2.2.1 K-Medoids

The K-Medoids algorithm is a clustering method used to group (n) objects into several (k) clusters [21], [22]. In this method, the objects selected as representatives of each cluster are known as medoids [21], [22]. K-Medoids has the characteristic that the cluster center is between its data points, and uses Euclidean Distance to calculate the distance between the object and the center (medoid), so that randomly selected objects tend to be similar to the medoid object [23]. K-Medoids is a method that is an extension of K-Means, where K-Medoids is specifically designed to overcome the weakness of K-Means which is vulnerable to the influence of outlier data [1], [24].

According to [25] outliers refer to observation objects that have very extreme values, either too low or too high, thus causing significant differences with other objects. K-Means, the previous method, is susceptible to the influence of outliers due to the presence of items with some very large values that can affect the distribution of data [26]. When using the

K-Means method, the average value (mean) is used as the cluster center to represent each cluster. However, in the K-Medoids method, objects (medoids) are used as cluster centers representing each cluster [2].

The following are the steps in calculating the K-Medoids algorithm, including [27]:

1. Determine the number of clusters (k) of the data.
2. Randomly select the first k medoids from the n data.
3. Calculate the Euclidean distance between each object and the medoids using the equation:

$$d(x, y) = |x - y| = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (2)$$

Where:

d : distance between x and y

x : cluster center data

y : data on the attribute

i : every data

n : amount of data

xi : data at the center of the i-th cluster

yi : data on each i-th data

4. Determine the closest medoid for each object and calculate its total distance.
5. Assign objects to the corresponding medoids to form temporary clusters.
6. Repeat steps 2 to 5 by replacing the medoids in the same way.
7. Calculate the total deviation (S) using the formula $S = b - a$, where:
 - a is the total number of closest distances between objects and medoids in the previous iteration or iteration 1.
 - b is the total number of closest distances between objects and the medoid at the current iteration or iteration 2.
8. Evaluate the total deviation (S):
 - If $S > 0$, the clustering process stops and the cluster member for each medoid is determined.
 - If $S < 0$, swap objects with other data to form a new set of medoids. Repeat the iteration until $S > 0$ is obtained.

2.2.2.2 K-Means

K-Means clustering is a cluster analysis technique that divides objects into k groups based on the nearest mean, with k as an input parameter that determines the number of clusters generated [12]. This algorithm accepts data as input, as well as a value of k that determines the number of clusters to be formed [1]. Each data or object will be placed into one of the k clusters, and each cluster has a centroid that represents the cluster [1]. The advantages of K-Means include ease of implementation, relatively faster calculation and computation time, and high popularity [1]. The effectiveness of this algorithm depends on the initial position of the centroid, where after the number k is calculated, the data points are assigned to the nearest centroid based on the Euclidean distance to form clusters, and after cluster formation, the centroid is recalculated by taking the average of the clusters, and this iteration continues until there is no change in the position of the centroid [28].

The steps in K-Means clustering are as follows [29][26]:

1. Select object k as the initial center of each cluster.
2. Calculate the Euclidean distance using equation (2) between each object and each centroid.
3. Assign each object to the cluster that is closest to it, based on the distance between the object and the centroid.
4. Update the centroid position by calculating the average value of the objects in each cluster.
5. Repeating steps 2-4 until there is no further change in the position of the centroid or cluster center.
6. The analysis is complete when there is no more change in the position of the centroid or cluster center.

2.2.3 Cluster Validation

This validation test stage is conducted to compare the performance between the two clustering methods using the DBI value and execution time.

2.2.3.1 Davies-Bouldin Index

Davies-Bouldin Index (DBI) is a metric used to evaluate the results of clustering algorithms [12][27]. This metric was introduced by David L. Davies and Donald W. Bouldin in 1979 [27][12]. DBI is used to determine whether a clustering scheme has optimal quality and a clustering scheme that is considered optimal is one that has a minimal DBI value [12]. To get the DBI value, the following steps must be applied, namely [30]:

1. Counting Sum Of Square Within Cluster (SSW)
2. Used to find the cohesion value in the i -th cluster which can be seen in the equation (3).

$$SSW = \frac{1}{m_i} \sum_{j=1}^{m_i} d(x_j, c_i) \quad (3)$$

Description:

m_i : Number of data in the i -th cluster

x_j : Data in clusters

c_i : The center point of the i -th cluster

$d(x_j, c_i)$: Euclidean distance of each data to the center point

3. Counting Sum Of Square Between-Cluster (SSB)

Used to find the separation between clusters whose equation can be seen at (4).

$$SSB_{ij} = d(C_i, C_j) \quad (4)$$

Description:

C_i : First cluster

C_j : Other clusters

$d(C_i, C_j)$: Euclidean distance from one center point to another

4. Counting Ratio

5. Used to get the comparison value between the i-th cluster and the j-th cluster. The ratio calculation can be seen in the equation (5).

$$R_{i,j} = \frac{SSW_i + SSW_j}{SSB_{i,j}} \tag{5}$$

- Description:
- $R_{i,j}$

: Ratio between clusters
- SSW_i

: The i-th cluster
- SSW_j

: The j-th cluster
- $SSB_{i,j}$

: Separation of i-th and j-th clusters

6. Counting DBI

Used to get the DBI value, the equation can be seen at (6).

$$DBI = \frac{1}{k} \sum_{i=1}^k \max_{i \neq j} (R_{i,j}) \tag{6}$$

- Description:
- k

: Existing clusters
- $R_{i,j}$

: Ratio between clusters i and j
- $\max$

: Find the largest inter-cluster ratio

2.2.3.2 Execution Time

Execution Time is an important measure in data analysis and computing that refers to the time taken by a program or algorithm to complete a specific task or operation [31]. The main purpose of measuring execution time is to evaluate the performance of a clustering algorithm and identify whether it can deal with data of different sizes and complexities [31][15]. The faster the algorithm can complete its task, the better because shorter execution time means that the algorithm is more efficient in processing data [14]. Algorithms with lower execution time may be more suitable for large or complex data, while algorithms with higher execution time may be more suitable for smaller or simpler data [14].

2.2.4 Scoring Variabel RFM

The Recency, Frequency, and Monetary (RFM) model is used to classify and categorize customers into segments that require special treatment [1]. RFM models require historical data such as purchase history, thus enabling implementation in the marketing space [1]. RFM analysis consists of three main components, namely recency, frequency, and monetary [5][8]. Recency is a variable used to assess customer value based on the time that has passed since the customer last made a purchase transaction [8][5]. Frequency is a variable used to assess customer value based on the number of transactions made by customers in a certain period of time [5]. Monetary variables are used to assess customers based on the amount of money that has been spent by customers during a period of time [5].

This stage aims to give a score to each RFM variable based on the grouping results. RFM scores will be used as customer data that will be analyzed at a later stage. There are two score ranges used, namely scores 1-3 and scores 1-5. The use of these two score ranges

makes it possible to obtain more detailed customer segmentation analysis results [18]. Scores 1-3 are used for customer segmentation with fewer category levels [18], while scores 1-5 are used for more in-depth and detailed segmentation [17]. Thus, the use of two score ranges provides flexibility in analyzing and understanding customer patterns and characteristics at various levels.

3 Results and Discussion

The online retail dataset consists of eight attributes, namely Invoice No, Stock Code, Description, Quantity, Invoice Date, Unit Price, Customer ID, and Country, with a total of 64451 transactions. Prior to analysis, the data is pre-processed on the Quantity and Unit Price attributes by removing data rows that contain negative values.

3.1 Pre-processing

In the preprocessing stage, data transformation is carried out by adding new variables generated from existing attributes, namely quantity multiplied by unitprice. This new variable will be used for monetary analysis in the RFM model. After the attribute amount is formed, then create a pivot table which has a function to group data according to customer id. Then create a max of invoice date variable which functions to find out the date of the last purchase for each customer which will then be useful for analyzing the recency variable in RFM analysis.

In addition to creating a max of invoice date variable, a count of invoiceNo variable is also created by using the count distinct function in excel which functions so that the same invoice data on the same day for each customer is made into one invoice, so there will be no data redundancy. This variable creation is useful for analyzing the frequency variable in RFM analysis. The following is a sample of online retail transaction data that has been transformed, found in Table 1 and Table 2.

Table 1. Transformed data

Customer ID	Max of InvoiceDate	Sum of Amount	Distinct Count of InvoiceNo	Day Since
12347	07/12/2011 15:52	4310	7	2
12348	25/09/2011 13:13	1797,24	4	75
12349	21/11/2011 09:51	1757,55	1	18
12350	02/02/2011 16:01	334,4	1	310
12352	03/11/2011 14:37	1545,41	11	36

Table 2. Final result of data transformation

Customer ID	Recency <i>y</i>	Frequency <i>y</i>	Monetary <i>y</i>
12347	2	7	4310
12348	75	4	1797,24
12349	18	1	1757,55
12350	310	1	334,4

Customer ID	Recency y	Frequency y	Monetary y
12352	36	11	1545,41
12353	204	1	89

After passing the transformation stage, the data is normalized using z-score normalization. The use of z-score normalization for data normalization is because this normalization method is more recommended when data has outliers because it is more resistant to the impact of outliers [32]. Z-score normalization transforms the data into a normal distribution with mean 0 and standard deviation 1 [33]. Therefore, outliers that are far from the mean will approach the value of -3 or 3 in a normal distribution [34]. This makes z-score normalization more effective in handling outliers than minmaxscaler which uses the minimum and maximum values of the data in normalization [33]. The following is a sample of normalized online retail transaction data, found in Fig. 2.

	Frequency	Monetary	Recency
0	-0.060026	2.660264	2.329641
1	0.029638	0.065921	-0.900392
2	-0.045165	-0.023535	-0.170384
3	-0.024359	-0.024948	-0.740390
4	-0.052100	-0.075613	2.179640

Fig. 2. Normalized data

3.2 Elbow Method

Furthermore, to determine the optimal value of k clusters, the elbow method is used using a python program before entering the clustering stage. The result of determining the optimal k value is shown in Fig. 3.

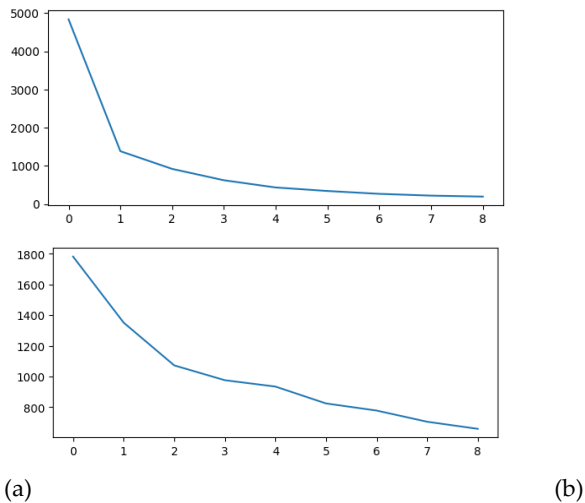


Fig. 3. Graph visualization of elbow method: (a) K-Means and (b) K-Medoids

Based on the results of the k value graph with the elbow method, it shows that the optimal number of clusters (k) in the K-Means method is 3, while in the K-Medoids method is 5. Although on the graph there seems to be an elbow at the value of k = 1, but when k = 1, all data will be merged into a single group regardless of the existing variations. So k will be set at 3 because with k=3, the clusters are able to represent the variations and patterns in the data better than with only 1 cluster.

3.3 Clustering with K-Medoids and K-Means Methods

In the clustering process using K-Medoids and K-Means, the data is grouped based on the similarity of the characteristics of each customer with the clusters formed are 5 clusters in K-Medoids, namely clusters 0, 1, 2, 3, and 4, while in K-Means the clusters formed are 3, namely clusters 0, 1, and 2. The results of data clustering on the online retail dataset can be seen in Fig. 4.

The clustering results using K-Means with parameter k=3 and K-Medoids with parameter k=5 can be seen. Visualization of data distribution shows the difference in handling outlier data between the two methods.

In the K-Means method, outlier data tends to be in the same cluster. This is because K-Means uses the average (centroid) of the cluster as the cluster representation, so extreme values of outliers can affect the cluster center and direct the cluster to include outlier data. This may cause the clusters to be distorted by the presence of outliers.

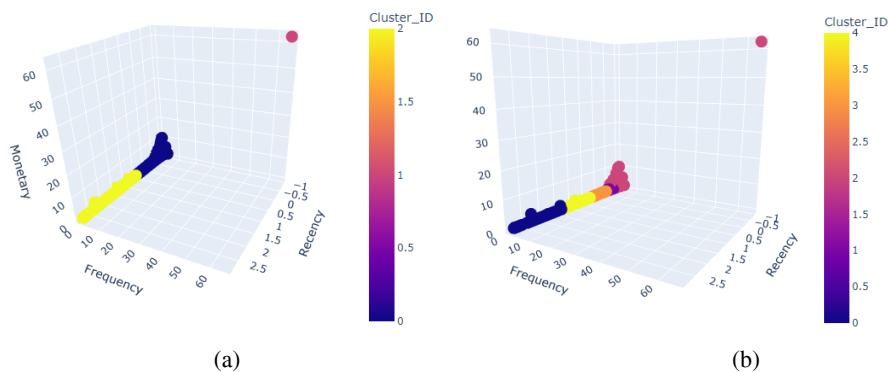


Fig. 4. Clustering graph visualization: (a) K-Means and (b) K-Medoids

On the other hand, in the K-Medoids method, outlier data can be in a different cluster to the majority of the data. K-Medoids uses the medoid, which is the actual data in the cluster, as a representation of the cluster. Since the medoids are not influenced by the extreme values of the outliers, K-Medoids is more resilient to the impact of outliers and tends to leave the outliers in clusters that match their characteristics.

3.4 Validation Test

The clustering validation test process looks at two main aspects, namely execution time and Davies-Bouldin Index (DBI) value. Execution time measures how fast the clustering algorithm can complete its task, while DBI is used to measure the quality of the resulting clustering by considering an important aspect in data clustering, namely density. DBI evaluates how well the clusters are formed, with lower values indicating better clustering.

Table 3. Execution Time Comparison Results

Algorithm	Execution time
K-Medoids	2.4295 sec
K-Means	0.0960 sec

Based on Table 3, it can be seen that in the clustering process, K-Means is faster than K-Medoids. This is because K-Means uses the mean as the cluster center, which allows for more efficient calculations than K-Medoids which uses medoids [22]. In conclusion, in the context of this online retail dataset, the K-Means algorithm provides more efficient clustering time than K-Medoids. This happens because K-Means has a computational complexity that tends to be lower than K-Medoids, which means it can cope with larger datasets more efficiently as stated in research [1] which states that the calculation and computation time in K-Means is relatively faster.

Table 4. DBI value comparison results

Algorithm	Davies-Bouldin Index	
	Manhattan	Euclidean
K-Medoids	0.9688	0.8942
K-Means	-	0.2962

Based on the above, the DBI value of the K-Medoids algorithm is 0.8942 and K-Means is 0.2962 using euclidean distance, while the DBI value result for the K-Medoids algorithm using manhattan distance is 0.9688. Based on the consideration of the DBI results shown in Table 4, the distance used in the K-Medoids algorithm is Euclidean distance because it has a smaller DBI value than using Manhattan distance.

The best result based on the DBI value is found in the K-Means algorithm with a DBI of 0.2962, the value is considered good because it is the smallest value obtained. In conclusion, K-Means can provide a lower value because it prioritizes density rather than overcoming the distance between clusters which may be greater due to the influence of outliers as has been stated in research [12], [27] that DBI measures how well the grouping is by looking at how close the center of each group is and how separate the groups are, DBI is lower if the group centers are closer to each other and the distance between the group centers and other groups is greater.

3.5 RFM Variable Scoring

In the RFM variable scoring step, the data that has been grouped will be given a score for each variable. This variable score can be in two different ranges, namely score ranges 1-3 and 1-5. For scoring with a range of 1-3 on RFM variables, quantiles with a value range of 0.25, 0.5, and 0.75 are used. Meanwhile, for scoring with a range of 1-5 on the RFM variable, quantiles with a range of values of 0.2, 0.4, 0.6, 0.8, and 1 are used. The results of scoring the RFM variable against the K-Means and K-Medoids methods can be seen in Table 5, Table 6, Table 7, and Table 8.

Table 5. RFM 1-3 scoring results on K-Means

Custo merID	rece ncy	frequ ency	monetary _value	Cluste r_ID	R_S core	F_Sc ore	M_S core	RFMS core
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12346.0	325	1	77183,6	2	1	1	3	113
12347.0	2	182	4310	0	3	3	3	333
12348.0	75	31	1797,24	0	2	2	3	223
12349.0	18	73	1757,55	0	2	2	3	223

Table 6. RFM scoring results 1-5 on K-Means

Custo merID	rece ncy	frequ ency	monetary _value	Cluste r_ID	R_S core	F_Sc ore	M_S core	RFMS core
12346.0	325	1	77183,6	2	1	1	5	115
12347.0	2	182	4310	0	5	5	5	555
12348.0	75	31	1797,24	0	2	3	4	234
12349.0	18	73	1757,55	0	4	4	4	444

Table 7. RFM scoring results 1-3 on K-Medoids

Custo merID	rece ncy	frequ ency	monetary _value	Cluste r_ID	R_S core	F_Sc ore	M_S core	RFMS core
12346.0	325	1	77183,6	0	1	1	3	113
12347.0	2	182	4310	2	3	3	3	333
12348.0	75	31	1797,24	3	2	2	3	223
12349.0	18	73	1757,55	2	2	2	3	223

Table 8. RFM scoring results 1-5 on K-Medoids

Custo merID	rece ncy	frequ ency	monetary _value	Cluste r_ID	R_S core	F_Sc ore	M_S core	RFMS core
12346.0	325	1	77183,6	0	1	1	5	115
12347.0	2	182	4310	2	5	5	5	555
12348.0	75	31	1797,24	3	2	3	4	234
12349.0	18	73	1757,55	2	4	4	4	444

3.6 Analysis of Clustering Results

At the stage of analyzing the results of data clustering, the aim is to identify customer groups and determine the characteristics shared by each customer group in each segment based on the number of clusters formed. The results of this analysis guide companies in designing marketing strategies and provide information about the characteristics of customers in each group. To find out the type of customer, customer mapping can be done according to [35] for RFM range 1-5, for RFM ranges 1-5, while for RFM ranges 1-3, customer mapping is carried out according to [36][36][4].

The classification results based on RFM values with score representation show that in scoring 1-3 with the K-Means clustering method, segment 0 is in the category of "Potential Customer/ Golden Customer," segment 2 is in the category of "Lost Customer/ Dormant Customer," and segment 1 is in the category of "Superstar/ Core Customer." The average RFM score per segment can be seen in Table 9. In determining the customer characteristics

per segment, attention is paid to the majority of the highest RFM scores as a reference that will represent the segment.

Table 9. Average RFM score 1-3 per K-Means segment

Segm ent	Number of Customers	Average Value Per Segment			Sk or RF M	Customer Type
		R	F	M		
0	3269	2 (65,77 %)	2 (49,98 %)	2 (50,96 %)	22 2	Potential Customer/ Golden Customer
2	1069	1 (100%)	1 (48,46 %)	1 (48,46 %)	11 1	Superstar/ Core Customer
1	1	3 (100%)	3 (100%)	3 (100%)	33 3	Lost Customer/ Dormant Customer

The segmentation results using K-Medoids based on RFM scores ranging from 1-3 show five segments. Segment 0 is categorized as "Lost Customer/ Dormant Customer", segment 1 has the potential to become a "Potential Customer/ Golden Customer", segment 2 is a "Superstar/ Core Customer", segment 3 also has the potential to become a "Potential Customer/ Golden Customer", and segment 4 is in the "At Risk Customers/Occasional customer" category. However, analysis of the RFM results shows that the score range of 1-3 using K-Medoids is not able to break up many segments. This is due to the average value per segment, where segments 1 and 3 have the same average score of 222. Details of the average RFM score value per segment can be found in Table 10.

Table 10. Average RFM score 1-3 per K-Medoids Segment

Segm ent	Number of Customers	Average Value Per Segment			Sk or RF M	Customer Type
		R	F	M		
0	606	1 (100%)	1 (54,13 %)	1 (54,62 %)	11 1	Lost Customer/ Dormant Customer
1	1015	2 (100%)	2 (54,38 %)	2 (55,67 %)	22 2	Potential Customer/ Golden Customer
2	1263	3 (87,41 %)	3 (49,88 %)	3 (49,17 %)	33 3	Superstar/ Core Customer
3	866	2 (100%)	2 (57,16 %)	2 (60,39 %)	22 2	Potential Customer/ Golden Customer
4	589	1 (81,32 %)	2 (54,16 %)	2 (53,48 %)	12 2	At Risk Customers/ Occasional customer

Then, in addition to using a score range of 1-3, a score range of 1-5 is also used in determining customer segmentation. The segmentation results with clustering using the K-Means method based on the RFM score range 1-5 show that segment 0 is in the "Champion" category, segment 2 is in the "Lost" category, and segment 1 is in the "Champion" category as well. However, analysis of the RFM results shows that the score range of 1-5 using K-Means is not able to break up many segments. This is due to the average value per segment, where segments 0 and 1 have the same customer type, which is Champion. The average value of RFM score per segment can be seen in Table 11.

Table 11. Average RFM score 1-5 per K-Means segment

Segme nt	Number of Customers	Average Value Per Segment			Sko r RF M	Customer Type
		R	F	M		
0	3269	4 (26,94%)	5 (25,45%)	5 (25,54%)	455	Champion
2	1069	1 (80,54%)	1 (41,72%)	1 (39,66%)	111	Lost
1	1	5 (100%)	5 (100%)	5 (100%)	555	Champion

Then, for K-Medoids clustering, a customer segmentation analysis was conducted based on RFM scores ranging from 1-5. The analysis results show that segment 0 is in the category of "Lost," segment 1 is in the category of "Loyal Customer," segment 2 is in the category of "Champion," segment 3 is in the category of "At Risk," and segment 4 is in the category of "Hibernating." Based on the results of this analysis, it can be concluded that the use of a score range of 1-5 using K-Medoids clustering is able to solve many segments better. The average value of the RFM score per segment can be seen in Table 12.

Table 12. Average RFM score 1-5 per segment K-Medoids

Segme nt	Number of Customers	Average Value Per Segment			Sko r RF M	Customer Type
		R	F	M		
0	606	1 (100%)	1 (46,86%)	1 (45,38%)	111	Lost
1	1015	3 (52,02%)	4 (26,11%)	4 (24,83%)	344	Loyal Customer
2	1263	5 (68,80%)	5 (42,44%)	5 (42,20%)	555	Champion
3	866	2 (61,43%)	3 (25,64%)	3 (25,87%)	233	At Risk
4	589	2 (56,71%)	1 (33,28%)	1 (30,73%)	211	Hibernating

After analyzing the results of grouping and scoring RFM variables, the characteristics and types of customers per segment were successfully identified. With this information, the

next step is to develop appropriate marketing strategy recommendations. The following are marketing strategies that can be recommended:

1. *Champion/Supersstar/Core Customer*; This type of customer is one who has completed a purchase recently. They purchase frequently and transact the most. They have the highest RFM score. For this type of customer, a marketing strategy that can be recommended is to give awards as a form of appreciation for their loyalty. These customers have the potential to be early adopters of new products, meaning they tend to be interested and try the latest products as soon as possible. In addition, they also have a significant role as brand ambassadors, as they tend to share their positive experiences with others and recommend products or services to friends and family. Therefore, providing incentives or reward programs to these customers can strengthen their attachment to the brand and encourage more people to try our company's products or services.
2. *Loyal Customer*; Customers with this type are customers who buy frequently and spend a lot of money at the company, but have a lower recency range than customers with the champion type. A marketing strategy that can be recommended is Upsell. Upsell is an approach to offer higher value products or services to existing customers. In this strategy, offer customers goods or services that have additional features or superior quality, which will likely increase their satisfaction. In addition, it is also important to ask for reviews from customers. Reviews from customers can help boost the confidence of potential new buyers and give an idea of the extent to which customers are satisfied with the products or services they receive.
3. *Potential Loyalist/Potential/Customer/ Golden Customer*; This type of customer is a customer who has recently spent a considerable amount of money at this company more than once. A marketing strategy that can be a recommendation is to invite customers to join the loyalty program and provide additional product recommendations, companies can strengthen bonds with customers, increase retention, and encourage further purchases, thus creating a mutually beneficial long-term relationship.
4. *At Risk/At Risk Customers/Occasional customer*; customers with this type are customers who completed large and frequent purchases a long time ago, but it has been quite a while since the customer has made a transaction. A marketing strategy that can be a recommendation is to send personalized emails with special offers and share useful resources, companies can create positive interactions with customers, increase engagement, and strengthen long-term relationships with them.
5. *Hibernating*; This type of customer is one whose last purchase was a long time ago. The number of orders is small and the amount spent is low. Marketing strategies that can be recommended are offering relevant products and providing special discounts and recreating brand value, companies can attract customer attention, build trust, and strengthen customer loyalty.
6. *Lost/Lost Customer/Dormant Customer*; This type of customer is the one with the lowest value in terms of recency, frequency, and value for money. Try to reactivate them with a personalized campaign. If it doesn't work, just ignore it.

4 Conclusion

This research performs customer segmentation using two clustering methods, namely K-Means and K-Medoids, by utilizing the RFM model (Recency, Frequency, Monetary). The elbow method was used to determine the best number of clusters for each algorithm. The results showed that K-Means produced 3 best clusters, while K-Medoids produced 5 best clusters.

Comparative analysis was performed using two clustering evaluation metrics, namely Davies-Bouldin Index (DBI) and execution time. The results show that K-Means has a better DBI value (0.2962) than K-Medoids (0.8942), indicating that K-Means provides better and more optimal clustering quality compared to K-Medoids. However, in the case of data that has outliers, it can be concluded that K-Means can provide a lower value because it prioritizes density rather than overcoming the distance between clusters that may be greater due to the influence of outliers based on the visualization results that have been done.

In addition, the execution time is also evaluated, and it is found that K-Means is faster (0.0960 seconds) than K-Medoids (2.4295 seconds). In the context of this online retail dataset, the K-Means algorithm provides more efficient clustering time than K-Medoids. This happens because K-Means has a computational complexity that tends to be lower than K-Medoids, which means it can cope with larger datasets more efficiently as stated in research [1] which states that the calculation and computation time on K-Means is relatively faster than K-Medoids.

Based on the segmentation results of each algorithm with the RFM model, K-Means produces customer levels with RFM score ranges 1-3 Potential Customer / Golden Customer, Lost Customer / Dormant Customer, and Superstar / Core Customer, while for the score range 1-5 is to produce Champion, Champion, and Lost. Then for the customer tiers produced by the K-Medoids method with RFM score range 1-5 are Lost, Loyal Customer, Champion, At Risk, and Hibernating, while in the RFM score range 1-3 it produces Lost Customer / Dormant Customer, Potential Customer / Golden Customer, Superstar / Core Customer, Potential Customer / Golden Customer, and At Risk Customers / Occasional customers.

Analysis of the RFM results shows that the score range of 1-5 using K-Means is not able to break up many segments. This is due to the average value per segment, where segments 0 and 1 have the same type of customer, namely Champion. And analysis in the score range 1-3 using K-Medoids is not able to break many segments. This is due to the average value per segment, where segments 1 and 3 have the same customer type, namely Potential Customer / Golden Customer.

In conclusion, the results show that K-Means performs better than K-Medoids in terms of clustering quality based on DBI value and faster clustering execution time. However, when facing outlier data, K-Medoids is more robust than K-Means. The clustering visualization results show that in the K-Means method, outlier data is clearly separated from non-outlier data. The clusters generated by K-Means look compact and focused on the non-outlier data, while the outlier data is far away from the main cluster. The visualization illustrates that K-Means tends to be sensitive by outlier data. As for the cluster groups generated by K-Medoids, it can be seen that the outlier data is more integrated with the main group.

The results of the comparison between 1-3 and 1-5 ranges show that K-Medoids is able to map different types of customers better in the range of 1-5, while K-Means is more effective in mapping different types of customers in the range of 1-3 on the online retail dataset that has been used. The EDA process on this dataset only involves the steps of finding outlier data and removing negative values, without going through any other steps.

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References

- [1] S. I. Murpratiwi, I. G. Agung Indrawan, and A. Aranta, J. Pendidik. Teknol. dan Kejuru., **Vol. 18, No. 2**, p. 152, (2021), doi: 10.23887/jptk-undiksha.v18i2.37426.
- [2] R. Siagian, P. Sirait, and A. Halim, Sistemasi, **Vol. 11, No. 2**, p. 260, (2022), doi: 10.32520/stmsi.v11i2.1337.
- [3] A. B. H. Kiat, Y. Azhar, and V. Rahmayanti, J. Repos., **Vol. 2, No. 7**, p. 945, (2020), doi: 10.22219/repositor.v2i7.973.
- [4] B. E. Adiana, I. Soesanti, and A. E. Permanasari, J. Terap. Teknol. Inf., **Vol. 2, No. 1**, pp. 23–32, (2018), doi: 10.21460/jutei.2018.21.76.
- [5] K. Z. Wijaya, A. Djunaidi, and F. Mahananto, J. Tek. ITS, **Vol. 10, No. 2**, (2021), doi: 10.12962/j23373539.v10i2.67707.
- [6] W. M. Kifti and W. Swaradana, J. Sci. Soc. Res., **Vol. 3, No. 1**, pp. 57–63, (2020).
- [7] N. Puspitasari, J. A. Widians, and N. B. Setiawan, J. Teknol. dan Sist. Komput., **Vol. 8, No. 2**, pp. 78–83, (2020), doi: 10.14710/jtsiskom.8.2.2020.78-83.
- [8] F. Marisa, S. S. S. Ahmad, Z. I. M. Yusof, Fachrudin, and T. M. A. Aziz, Int. J. Integr. Eng., **Vol. 11, No. 3**, pp. 169–180, (2019), doi: 10.30880/ijie.2019.11.03.018.
- [9] C. A. Sugianto, A. H. Rahayu, and A. Gusman, J. Inf. Technol., **Vol. 2, No. 2**, pp. 39–44, (2020), doi: 10.47292/joint.v2i2.30.
- [10] Z. Wu, L. Jin, J. Zhao, L. Jing, and L. Chen, Comput. Intell. Neurosci., **Vol. 2022**, (2022), doi: 10.1155/2022/9930613.
- [11] M. Aryuni, E. Didik Madyatmadja, and E. Miranda, “Customer Segmentation in XYZ Bank Using K-Means and K-Medoids Clustering,” in Proc. 2018 Int. Conf. Inf. Manag. Technol. ICIMTech 2018, No. September 2018, pp. 412–416, (2018), doi: 10.1109/ICIMTech.2018.8528086.
- [12] I. Kamila, U. Khairunnisa, and M. Mustakim, J. Ilm. Rekayasa dan Manaj. Sist. Inf., **Vol. 5, No. 1**, p. 119, (2019), doi: 10.24014/rmsi.v5i1.7381.
- [13] A. Wibowo and A. R. Handoko, J. Nas. Teknol. dan Sist. Inf., **Vol. 5, No. 1**, pp. 17–24, (2019).
- [14] E. Zuccarelli, “Performance Metrics in Machine Learning — Part 3: Clustering,” *Towards Data Science*, (2021).
- [15] K. J. Wong, “7 Evaluation Metrics for Clustering Algorithms,” *Towards Data Science*, (2022).
- [16] R. W. Sembiring Brahmana, F. A. Mohammed, and K. Chairuang, J. Ilm. Teknol. Inf., **Vol. 11, No. 1**, p. 32, (2020), doi: 10.24843/lkjiti.2020.v11.i01.p04.
- [17] A. Sheshasaayee and L. Logeshwari, “Implementation of Clustering Technique Based Rfm Analysis,” in 2018 2nd Int. Conf. Trends Electron. Informatics, no. lcoei, pp. 1166–1170, (2018).
- [18] A. Panaiteescu, “How to calculate the RFM score for each customer,” *Omniconvert*,

- (2023). <https://www.omniconvert.com/blog/rfm-score/>
- [19] A. Winarta and W. J. Kurniawan, J. Tek. Inform. Kaputama, **Vol. 5, No. 1**, pp. 113–119, (2021).
 - [20] N. T. Hartanti, J. Nas. Teknol. dan Sist. Inf., **Vol. 6, No. 2**, pp. 82–89, (2020), doi: 10.25077/teknosi.v6i2.2020.82-89.
 - [21] S. Nurlaela, A. Primajaya, and T. N. Padilah, I N F O R M a T I K a, **Vol. 12, No. 2**, p. 56, (2020), doi: 10.36723/juri.v12i2.234.
 - [22] G. B. Kaligis and Y. Sri, “Analisa Perbandingan Algoritma K-Means, K_Medoids, dan X-Means Untuk Pengelompokkan Kinerja Pegawai,” **Vol. 01**, pp. 179–193, (2022).
 - [23] A. Ayu, D. Sulistyawati, and M. Sadikin, “Penerapan Algoritma K - Medoids untuk Menentukan Segmentasi Pelanggan,” **Vol. 10**, pp. 516–526, (2021).
 - [24] A. Wibowo and A. R. Handoko, J. Teknol. Inf. dan Ilmu Komput., **Vol. 7, No. 3**, p. 573, (2020), doi: 10.25126/jtiik.2020702925.
 - [25] P. R. Fitrayana and D. R. S. Saputro, Prism. Pros. Semin. Nas. Mat., **Vol. 5**, pp. 721–725, 2022.
 - [26] Athifaturrofifah, R. Goejantoro, and D. Yuniarti, J. EKSPONENSIAL, **Vol. 10, No. 2**, pp. 143–152, (2019).
 - [27] I. W. Septiani, A. C. Fauzan, and M. M. Huda, J. Sist. Komput. dan Inform., **Vol. 3, No. 4**, p. 556, (2022), doi: 10.30865/json.v3i4.4055.
 - [28] T. Kansal, S. Bahuguna, V. Singh, and T. Choudhury, “*Customer Segmentation using K-means Clustering*,” in Proc. Int. Conf. Comput. Tech. Electron. Mech. Syst. CTEMS 2018, pp. 135–139, (2018), doi: 10.1109/CTEMS.2018.8769171.
 - [29] S. H. Shihab, S. Afroge, and S. Z. Mishu, “*RFM Based Market Segmentation Approach Using Advanced K means and Agglomerative Clustering: A Comparative Study*,” in 2019 Int. Conf. Electr. Comput. Commun. Eng., pp. 1–4, (2019).
 - [30] K. Amrulloh, “Comparison Between Davies-Bouldin Index and Silhouette Coefficient Evaluation Methods in Retail Store Sales Transaction Data Clusterization Using K-Medoids Algorithm,” (2022).
 - [31] D. Barba, “Execution Times in Python,” Towards Data Science, (2022). <https://towardsdatascience.com/execution-times-in-python-ed45ecc1bb4d>
 - [32] N. Fei, Y. Gao, Z. Lu, and T. Xiang, “*Z-Score Normalization, Hubness, and Few-Shot Learning*,” in Proc. IEEE Int. Conf. Comput. Vis., pp. 142–151, (2021), doi: 10.1109/ICCV48922.2021.00021.
 - [33] H. Henderi, IJIS Int. J. Informatics Inf. Syst., **Vol. 4, No. 1**, pp. 13–20, (2021), doi: 10.47738/ijis.v4i1.73.
 - [34] S. Kappal, London Journals Press, **Vol. 19, No. 4**, pp. 39–44, (2019).
 - [35] Connectif, “What Are RFM Scores and How To Calculate Them,” Connectif, 2022.
 - [36] P. Makhija, “RFM analysis for Customer Segmentation,” CleverTap, 2021.

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